Quantum-Ready Cybersecurity for Future-Safe Banking

PNB Cyber Security Hackathon 2026

Project: *Quantum-Proof Systems Scanner*

Team: *Neural Ninjas*

Team Members

Kamlesh Malviya | Prince Lavania

Team Members

Vijay Antony Rinish V

Indian Institute of Information Technology Kottayam

Mentor

Dr. Syamala S , Ass. Prof CSE

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The Emerging Quantum Threat

Banks rely on cryptography to protect critical financial data. However, future quantum computers may break traditional encryption algorithms.

Attack Scenario: Harvest Now – Decrypt Later (HNDL)

Attackers capture encrypted traffic today and decrypt it once quantum computers become available.

Challenges

Hidden Cryptographic Assets

Unknown cryptographic assets across infrastructure

Limited Visibility

Lack of visibility into TLS configurations

No PQC Readiness

No readiness assessment for Post-Quantum Cryptography (PQC)

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Proposed Solution

NEURAL NINJAS PQC SECURITY PLATFORM

A centralized cybersecurity platform that:

Discovers Public Facing Assets

Identifies all public-facing domains and subdomains across infrastructure.

Scans Cryptographic Configurations

Extracts TLS versions, cipher suites, and certificate details.

Builds a CBOM

Builds a Cryptographic Bill of Materials (CBOM) for every asset.

Evaluates PQC Readiness

Evaluates Post-Quantum Cryptography readiness across the estate.

Enterprise Security Analytics

Provides enterprise security analytics and risk scoring.

Automated Security Reports

Generates automated security reports for decision makers.

Goal: Enable banks to identify and migrate vulnerable cryptographic systems before quantum threats emerge.

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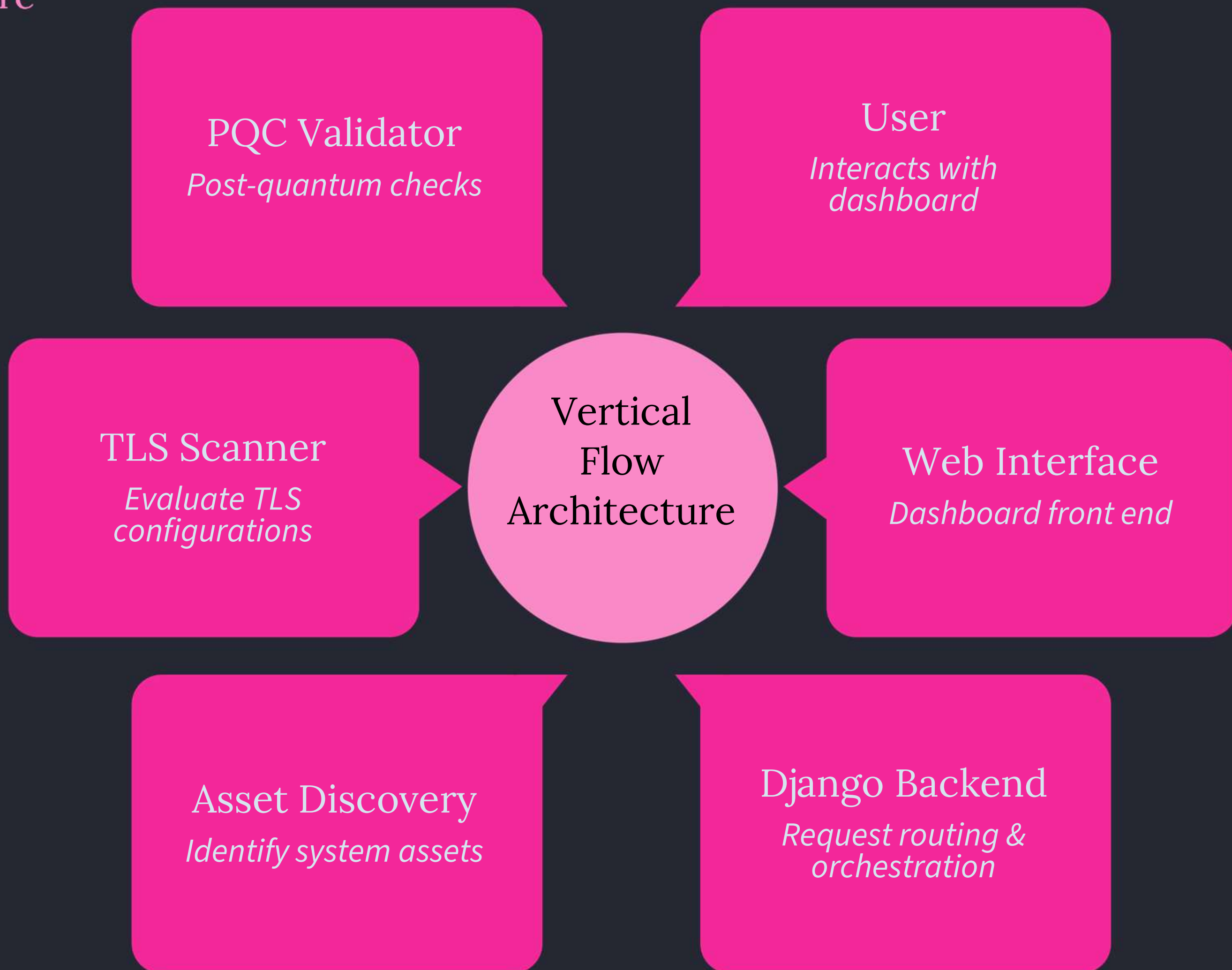
System Architecture

Create this diagram in draw.io or Lucidchart.



The architecture flows from the user-facing web interface through the Django backend, distributing work across six specialised modules before persisting results to the database and surfacing insights on the Security Analytics Dashboard.

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Platform Workflow

1. Asset Discovery

2. TLS Scan

3. Cryptographic Inventory

4. PQC Risk Evaluation

5. Security Analytics

6. Reporting

Simple pipeline: *Discover → Scan → Analyse → Evaluate PQC → Dashboard → Reports*

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Core Modules

Asset Discovery

Identify public facing domains and subdomains.

TLS Scanner

Extract TLS version, cipher suites, certificates.

CBOM Generator

Build Cryptographic Bill of Materials for each asset.

PQC Validator

Evaluate readiness for post-quantum algorithms.

Security Analytics

Calculate cyber risk rating and posture metrics.

Reporting Engine

Generate security reports for decision makers.

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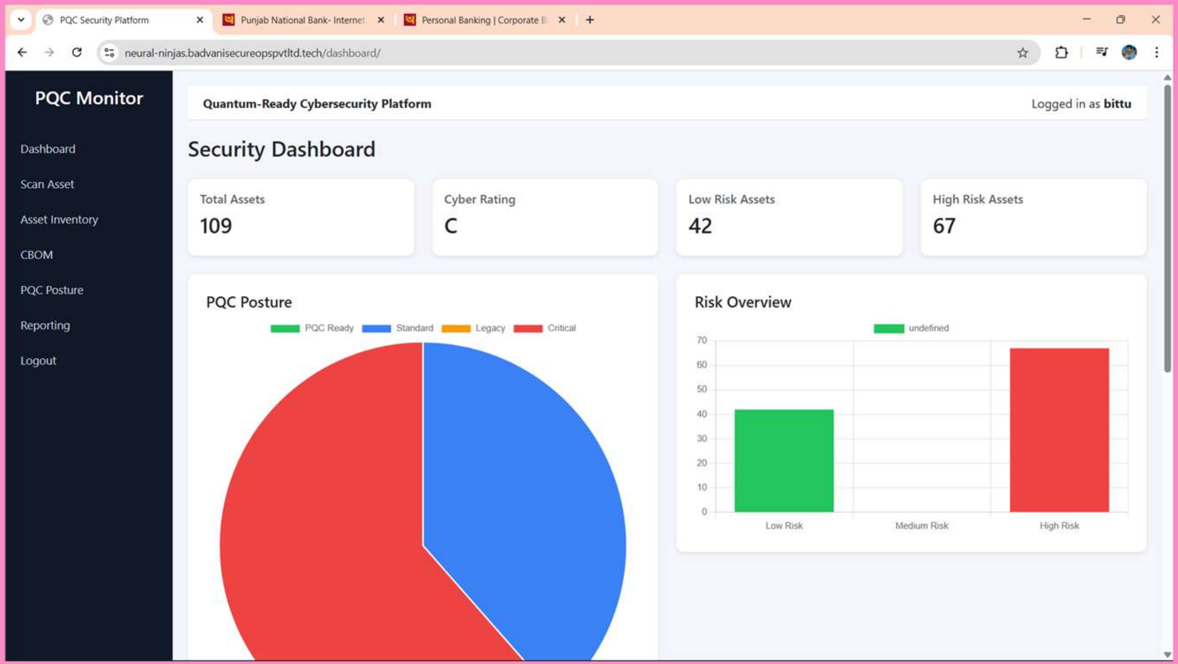
Security Analytics Dashboard

The dashboard surfaces the following key metrics:

Graphs displayed:

- *PQC readiness distribution*
- *Risk distribution*
- *Cyber rating*

Dashboard Preview



Total Scanned Assets

PQC Readiness Classification

Cyber Risk Score

Certificate Expiry Alerts

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CBOM Generation

The platform automatically builds a cryptographic inventory. This provides complete visibility of cryptographic assets across infrastructure.

Field	*Description*
TLS Version	*Protocol version in use for each asset*
Cipher Suite	*Negotiated cipher suite for the connection*
Signature Algorithm	*Algorithm used to sign the certificate*
Key Size	*Bit length of the cryptographic key*
Certificate Issuer	*Certificate Authority that issued the cert*
Certificate Validity	*Expiry date and validity window*
PQC Readiness Status	*Whether the asset is quantum-safe*
Risk Score	*Calculated risk rating for the asset*

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Security Features

Authentication System

Secure Login

Hardened login flow protecting platform access.

Two-Factor Authentication (2FA)

Additional verification layer for all users.

Risk Detection

- *TLS downgrade detection*
- *Cipher suite changes*
- *Certificate expiry alerts*

Security Change Monitoring

Alerts triggered when cryptographic configuration changes.

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Live Prototype

Live platform demo: https://neural-ninjas.badvanisecureopspvtltd.tech/

Features Implemented

- Asset Scanning
- CBOM Generation
- PQC Analytics Dashboard
- Security Reporting System
- Two Factor Authentication

The platform is fully deployed on a production server using Django and Gunicorn.

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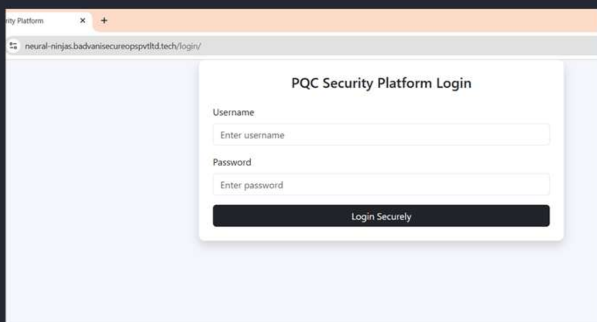
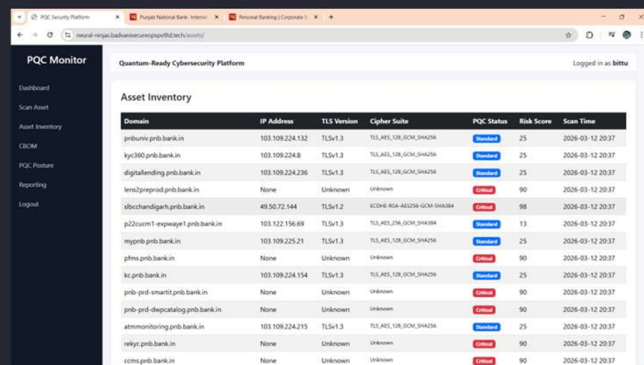
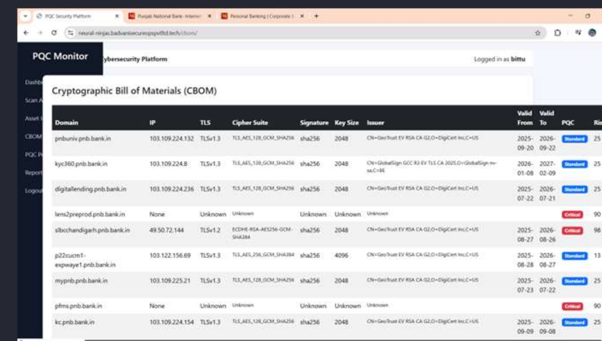
Prototype Screens

Insert 3–4 cropped UI images







*Do **NOT insert full browser screenshots**. Crop the relevant UI section.*

Examples include:

- *Dashboard*
- *Scan centre*
- *CBOM page*
- *Reporting page*

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Future Enhancements

Planned Improvements

- *Integration with enterprise asset management systems*
- *Automated scheduled scanning*
- *PQC migration recommendation engine*
- *Integration with Nmap and vulnerability scanners*
- *Real-time security alerts*
- *Support for NIST standardised PQC algorithms (Kyber, Dilithium)*

Impact

Helps banks transition safely into the ***post-quantum security era.***

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Thank You

Neural Ninjas – IIIT Kottayam

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